Week 7: SQL Query Implementation for E-Commerce Database

Scenario:

The online commerce platform maintains integrated records spanning customer accounts, catalog products, order transactions, and warehouse stock levels. Store administrators require targeted SQL queries to extract critical operational insights, verify transactions, and perform foundational business intelligence.

Students will write and execute SQL queries to search, filter, order, and compile analytical reports leveraging the normalized E-Commerce Database schema.

Tasks to be Done:

1. Perform Basic SQL Queries

Concept:

The SQL SELECT statement serves as the core mechanism for querying and projecting records from database tables.

1. Display all customer details.

```
SQL> SELECT * FROM CUSTOMERS;

CUST_ID  FIRST_NAME      LAST_NAME       EMAIL_ADDR                    CONTACT_NO JOIN_DATE
-------- --------------- --------------- ----------------------------- ---------- ---------
CUST0001 Arjun           Kumar           arjun.kumar@gmail.com         9876543210 15-JAN-24
CUST0002 Priya           Sharma          priya.sharma@gmail.com        9123456780 28-MAR-24
CUST0003 Rahul           Menon           rahul.menon@gmail.com         9988776655 10-JUN-24

CUST_ID  FIRST_NAME      LAST_NAME       EMAIL_ADDR                    CONTACT_NO JOIN_DATE
-------- --------------- --------------- ----------------------------- ---------- ---------
CUST0004 Ananya          Iyer            ananya.iyer@gmail.com         9876123450 22-AUG-24
CUST0005 Vikram          Reddy           vikram.reddy@gmail.com        9012345678 05-NOV-24
CUST0006 Meera           Krishnan        meera.krishnan@gmail.com      9345678120 18-JAN-25

CUST_ID  FIRST_NAME      LAST_NAME       EMAIL_ADDR                    CONTACT_NO JOIN_DATE
-------- --------------- --------------- ----------------------------- ---------- ---------
CUST0007 Suresh          Nair            suresh.nair@gmail.com         9567891234 12-MAR-25
CUST0008 Kavya           Rajan           kavya.rajan@gmail.com         9789012345 25-MAY-25
CUST0009 Karthik         Subramanian     karthik.subramanian@gmail.com 8899776655 09-JUL-25

CUST_ID  FIRST_NAME      LAST_NAME       EMAIL_ADDR                    CONTACT_NO JOIN_DATE
-------- --------------- --------------- ----------------------------- ---------- ---------
CUST0010 Divya           Pillai          divya.pillai@gmail.com        9098765432 30-SEP-25
CUST0011 Aditya          Verma           aditya.verma@gmail.com        9876501234 14-DEC-25
CUST0012 Sneha           Patel           sneha.patel@gmail.com         9765432109 20-FEB-26

CUST_ID  FIRST_NAME      LAST_NAME       EMAIL_ADDR                    CONTACT_NO JOIN_DATE
-------- --------------- --------------- ----------------------------- ---------- ---------
CUST0013 Rohan           Bose            rohan.bose@gmail.com          9654321098 17-APR-26
CUST0014 Lakshmi         Venkatesan      lakshmi.venkatesan@gmail.com  9543210987 08-JUL-26
CUST0015 Nithin          Murugan         nithin.murugan@gmail.com      9432109876 03-OCT-26

15 rows selected.
```

2. Display all available products.

```
SQL> SELECT * FROM PRODUCTS;

PROD_ID ITEM_NAME                     CAT_REF UNIT_PRICE QTY_IN_STOCK
------- ----------------------------- ------- ---------- ------------
P000001 Laptop                        CAT001       55000           25
P000002 Wireless Mouse                CAT002         850          100
P000003 Mechanical Keyboard           CAT002        2500           60

PROD_ID ITEM_NAME                     CAT_REF UNIT_PRICE QTY_IN_STOCK
------- ----------------------------- ------- ---------- ------------
P000004 Bluetooth Speaker             CAT003        3200           45
P000005 Smartphone                    CAT004       28000           35
P000006 USB Cable                     CAT005         450          150

PROD_ID ITEM_NAME                     CAT_REF UNIT_PRICE QTY_IN_STOCK
------- ----------------------------- ------- ---------- ------------
P000007 Power Bank                    CAT005        1800           75
P000008 Smart Watch                   CAT006        6500           40
P000009 Headphones                    CAT003        4200           55

PROD_ID ITEM_NAME                     CAT_REF UNIT_PRICE QTY_IN_STOCK
------- ----------------------------- ------- ---------- ------------
P000010 Tablet                        CAT004       22000           30
P000011 Web Camera                    CAT007        2800           50
P000012 External Hard Disk            CAT008        5200           35

PROD_ID ITEM_NAME                     CAT_REF UNIT_PRICE QTY_IN_STOCK
------- ----------------------------- ------- ---------- ------------
P000013 Gaming Monitor                CAT009       18500           20
P000014 Laptop Stand                  CAT010        1500           80
P000015 Wireless Earbuds              CAT003        3500           65

15 rows selected.
```

3. Retrieve product names and prices only.

```
SQL> SELECT ITEM_NAME, UNIT_PRICE FROM PRODUCTS;

ITEM_NAME                                          UNIT_PRICE
-------------------------------------------------- ----------
Laptop                                                  55000
Wireless Mouse                                            850
Mechanical Keyboard                                      2500
Bluetooth Speaker                                        3200
Smartphone                                              28000
USB Cable                                                 450
Power Bank                                               1800
Smart Watch                                              6500
Headphones                                               4200
Tablet                                                  22000
Web Camera                                               2800
External Hard Disk                                       5200
Gaming Monitor                                          18500
Laptop Stand                                             1500
Wireless Earbuds                                         3500

15 rows selected.
```

4. Display all orders placed by customers.

```
SQL> SELECT CONCAT(C.FIRST_NAME, C.LAST_NAME) AS CUSTOMER_NAME,
  2  O.ORDER_REF, O.ORDER_TOTAL, O.ORDER_STATUS
  3  FROM CUSTOMERS C JOIN ORDERS O
  4  ON C.CUST_ID = O.CUST_ID;

CUSTOMER_NAME                 ORDER_REF ORDER_TOTAL ORDER_STATUS
----------------------------- --------- ----------- ------------
ArjunKumar                    ORD0001         55850 Delivered
PriyaSharma                   ORD0002          3350 Delivered
RahulMenon                    ORD0003         28000 Delivered
AnanyaIyer                    ORD0004          8300 Delivered
VikramReddy                   ORD0005         22000 Processing
MeeraKrishnan                 ORD0006          7000 Delivered
SureshNair                    ORD0007          6500 Shipped
KavyaRajan                    ORD0008          5700 Delivered
KarthikSubramanian            ORD0009         18500 Processing
DivyaPillai                   ORD0010          5200 Delivered
AdityaVerma                   ORD0011          7000 Shipped
SnehaPatel                    ORD0012          2450 Delivered
LakshmiVenkatesan             ORD0014         28000 Delivered
NithinMurugan                 ORD0015          8350 Processing

14 rows selected.
```

5. Retrieve payment details.

```
SQL> SELECT * FROM ORDER_PAYMENTS;

TXN_ID  ORDER_REF TXN_DATE  PAY_METHOD      TXN_STATUS AMOUNT_PAID
------- --------- --------- --------------- ---------- -----------
TXN0001 ORD0001   10-JAN-26 Credit Card     Completed        55850
TXN0002 ORD0002   25-JAN-26 UPI             Completed         3350
TXN0003 ORD0004   18-FEB-26 Debit Card      Completed         8300
TXN0004 ORD0006   15-MAR-26 Net Banking     Completed         7000
TXN0006 ORD0010   18-MAY-26 UPI             Completed         5200
TXN0007 ORD0012   20-JUN-26 Credit Card     Completed         2450
TXN0008 ORD0014   15-AUG-26 Debit Card      Completed        28000
TXN0009 ORD0014   14-AUG-26 Debit Card      Failed           28000
TXN0005 ORD0008   20-APR-26 COD             Completed         5700

9 rows selected.
```

2. Apply Filtering Conditions Using WHERE

1. Find products with price greater than ■5000.

```
SQL> SELECT * FROM PRODUCTS WHERE UNIT_PRICE > 5000;

PROD_ID ITEM_NAME                     CAT_REF UNIT_PRICE QTY_IN_STOCK
------- ----------------------------- ------- ---------- ------------
P000001 Laptop                        CAT001       55000           25
P000005 Smartphone                    CAT004       28000           35
P000008 Smart Watch                   CAT006        6500           40
P000010 Tablet                        CAT004       22000           30
P000012 External Hard Disk            CAT008        5200           35
P000013 Gaming Monitor                CAT009       18500           20

6 rows selected.
```

2. Display products available in stock.

```
SQL> SELECT * FROM INVENTORY WHERE AVAIL_STATE = 'Available';

INV_ID  PROD_ID VENDOR_ID QTY_ON_HAND AVAIL_STATE UPDATED_ON
------- ------- --------- ----------- ----------- ----------
INV0001 P000001 SEL0001            25 Available   15-JAN-26
INV0002 P000002 SEL0002           100 Available   20-JAN-26
INV0003 P000003 SEL0002            60 Available   25-JAN-26
INV0004 P000004 SEL0003            45 Available   05-FEB-26
INV0005 P000005 SEL0004            35 Available   15-FEB-26
INV0006 P000006 SEL0005           150 Available   01-MAR-26
INV0007 P000007 SEL0005            75 Available   10-MAR-26
INV0008 P000008 SEL0006            40 Available   20-MAR-26
INV0009 P000009 SEL0003            55 Available   05-APR-26
INV0010 P000010 SEL0004            30 Available   15-APR-26
INV0011 P000011 SEL0007            50 Available   01-MAY-26
INV0012 P000012 SEL0008            35 Available   15-MAY-26
INV0013 P000013 SEL0009            20 Available   10-JUN-26
INV0014 P000014 SEL0010            80 Available   15-JUL-26
INV0015 P000015 SEL0003            65 Available   20-SEP-26

15 rows selected.
```

3. Display completed orders.

```
SQL> SELECT * FROM ORDERS WHERE ORDER_STATUS = 'Delivered';

ORDER_REF CUST_ID  ORDER_DATE ORDER_TOTAL ORDER_STATUS
--------- -------- ---------- ----------- ------------
ORD0001   CUST0001 10-JAN-26        55850 Delivered
ORD0002   CUST0002 25-JAN-26         3350 Delivered
ORD0003   CUST0003 05-FEB-26        28000 Delivered
ORD0004   CUST0004 18-FEB-26         8300 Delivered
ORD0006   CUST0006 15-MAR-26         7000 Delivered
ORD0008   CUST0008 20-APR-26         5700 Delivered
ORD0010   CUST0010 18-MAY-26         5200 Delivered
ORD0012   CUST0012 20-JUN-26         2450 Delivered
ORD0014   CUST0014 15-AUG-26        28000 Delivered

9 rows selected.
```

4. Find products with rating above 4.

```
SQL> SELECT
  2  P.ITEM_NAME,
  3  ROUND(AVG(R.RATING_SCORE), 2) AS AVERAGE_RATING
  4  FROM PRODUCTS P
  5  JOIN PRODUCT_REVIEWS R
  6  ON P.PROD_ID = R.PROD_REF
  7  GROUP BY P.PROD_ID, P.ITEM_NAME
  8  HAVING AVG(R.RATING_SCORE) > 4
  9  ORDER BY AVERAGE_RATING DESC;

ITEM_NAME                                          AVERAGE_RATING
-------------------------------------------------- --------------
Smartphone                                                      5
Bluetooth Speaker                                               5
Laptop                                                          5
External Hard Disk                                            4.5
Headphones                                                    4.5
Wireless Earbuds                                              4.5
Smart Watch                                                   4.5

7 rows selected.
```

3. Sort Data Using ORDER BY

Tasks:

1. Display products from lowest to highest price.

```
SQL> SELECT * FROM PRODUCTS ORDER BY UNIT_PRICE;

PROD_ID ITEM_NAME                     CAT_REF UNIT_PRICE QTY_IN_STOCK
------- ----------------------------- ------- ---------- ------------
P000006 USB Cable                     CAT005         450          150
P000002 Wireless Mouse                CAT002         850          100
P000014 Laptop Stand                  CAT010        1500           80

PROD_ID ITEM_NAME                     CAT_REF UNIT_PRICE QTY_IN_STOCK
------- ----------------------------- ------- ---------- ------------
P000007 Power Bank                    CAT005        1800           75
P000003 Mechanical Keyboard           CAT002        2500           60
P000011 Web Camera                    CAT007        2800           50

PROD_ID ITEM_NAME                     CAT_REF UNIT_PRICE QTY_IN_STOCK
------- ----------------------------- ------- ---------- ------------
P000004 Bluetooth Speaker             CAT003        3200           45
P000015 Wireless Earbuds              CAT003        3500           65
P000009 Headphones                    CAT003        4200           55

PROD_ID ITEM_NAME                     CAT_REF UNIT_PRICE QTY_IN_STOCK
------- ----------------------------- ------- ---------- ------------
P000012 External Hard Disk            CAT008        5200           35
P000008 Smart Watch                   CAT006        6500           40
P000013 Gaming Monitor                CAT009       18500           20

PROD_ID ITEM_NAME                     CAT_REF UNIT_PRICE QTY_IN_STOCK
------- ----------------------------- ------- ---------- ------------
P000010 Tablet                        CAT004       22000           30
P000005 Smartphone                    CAT004       28000           35
P000001 Laptop                        CAT001       55000           25

15 rows selected.
```

2. Display customers alphabetically.

```
SQL> SELECT * FROM CUSTOMERS ORDER BY FIRST_NAME;

CUST_ID  FIRST_NAME      LAST_NAME       EMAIL_ADDR                    CONTACT_NO JOIN_DATE
-------- --------------- --------------- ----------------------------- ---------- ---------
CUST0011 Aditya          Verma           aditya.verma@gmail.com        9876501234 14-DEC-25
CUST0004 Ananya          Iyer            ananya.iyer@gmail.com         9876123450 22-AUG-24
CUST0001 Arjun           Kumar           arjun.kumar@gmail.com         9876543210 15-JAN-24

CUST_ID  FIRST_NAME      LAST_NAME       EMAIL_ADDR                    CONTACT_NO JOIN_DATE
-------- --------------- --------------- ----------------------------- ---------- ---------
CUST0010 Divya           Pillai          divya.pillai@gmail.com        9098765432 30-SEP-25
CUST0009 Karthik         Subramanian     karthik.subramanian@gmail.com 8899776655 09-JUL-25
CUST0008 Kavya           Rajan           kavya.rajan@gmail.com         9789012345 25-MAY-25

CUST_ID  FIRST_NAME      LAST_NAME       EMAIL_ADDR                    CONTACT_NO JOIN_DATE
-------- --------------- --------------- ----------------------------- ---------- ---------
CUST0014 Lakshmi         Venkatesan      lakshmi.venkatesan@gmail.com  9543210987 08-JUL-26
CUST0006 Meera           Krishnan        meera.krishnan@gmail.com      9345678120 18-JAN-25
CUST0015 Nithin          Murugan         nithin.murugan@gmail.com      9432109876 03-OCT-26

CUST_ID  FIRST_NAME      LAST_NAME       EMAIL_ADDR                    CONTACT_NO JOIN_DATE
-------- --------------- --------------- ----------------------------- ---------- ---------
CUST0002 Priya           Sharma          priya.sharma@gmail.com        9123456780 28-MAR-24
CUST0003 Rahul           Menon           rahul.menon@gmail.com         9988776655 10-JUN-24
CUST0013 Rohan           Bose            rohan.bose@gmail.com          9654321098 17-APR-26

CUST_ID  FIRST_NAME      LAST_NAME       EMAIL_ADDR                    CONTACT_NO JOIN_DATE
-------- --------------- --------------- ----------------------------- ---------- ---------
CUST0012 Sneha           Patel           sneha.patel@gmail.com         9765432109 20-FEB-26
CUST0007 Suresh          Nair            suresh.nair@gmail.com         9567891234 12-MAR-25
CUST0005 Vikram          Reddy           vikram.reddy@gmail.com        9012345678 05-NOV-24

15 rows selected.
```

3. Find top expensive products.

```
SQL> SELECT * FROM PRODUCTS ORDER BY UNIT_PRICE DESC
  2  FETCH FIRST 5 ROWS ONLY;

PROD_ID ITEM_NAME                     CAT_REF UNIT_PRICE QTY_IN_STOCK
------- ----------------------------- ------- ---------- ------------
P000001 Laptop                        CAT001       55000           25
P000005 Smartphone                    CAT004       28000           35
P000010 Tablet                        CAT004       22000           30
P000013 Gaming Monitor                CAT009       18500           20
P000008 Smart Watch                   CAT006        6500           40
```

4. Display latest orders first.

```
SQL> SELECT * FROM ORDERS ORDER BY ORDER_DATE DESC;

ORDER_REF CUST_ID  ORDER_DATE ORDER_TOTAL ORDER_STATUS
--------- -------- ---------- ----------- ------------
ORD0015   CUST0015 25-SEP-26         8350 Processing
ORD0014   CUST0014 15-AUG-26        28000 Delivered
ORD0012   CUST0012 20-JUN-26         2450 Delivered
ORD0011   CUST0011 03-JUN-26         7000 Shipped
ORD0010   CUST0010 18-MAY-26         5200 Delivered
ORD0009   CUST0009 05-MAY-26        18500 Processing
ORD0008   CUST0008 20-APR-26         5700 Delivered
ORD0007   CUST0007 01-APR-26         6500 Shipped
ORD0006   CUST0006 15-MAR-26         7000 Delivered
ORD0005   CUST0005 02-MAR-26        22000 Processing
ORD0004   CUST0004 18-FEB-26         8300 Delivered
ORD0003   CUST0003 05-FEB-26        28000 Delivered
ORD0002   CUST0002 25-JAN-26         3350 Delivered
ORD0001   CUST0001 10-JAN-26        55850 Delivered

14 rows selected.
```

4. Retrieve Unique Values Using DISTINCT

Tasks:

1. Display unique product categories.

```
SQL> SELECT DISTINCT CATEGORY_NAME FROM CATEGORIES ORDER BY CATEGORY_NAME;

CATEGORY_NAME
--------------------------------------------------
Audio Devices
Clothing
Computer Accessories
Electronics
Home
Jewellery
Laptop Accessories
Laptops
Mobile Devices
Monitors
Pharmacy
Power Accessories
Storage Devices
Wearables
Web Accessories

15 rows selected.
```

2. Find different payment methods used by customers.

```
SQL> SELECT DISTINCT PAY_METHOD AS PAYMENT_METHOD FROM ORDER_PAYMENTS;

PAYMENT_METHOD
------------------
Credit Card
UPI
Debit Card
Net Banking
COD
```

5. Search Products Based on Conditions

Tasks:

Write SQL queries to:

1. Find products within a specific price range. (Product Between 20,000 to 40,000)

```
SQL> SELECT * FROM PRODUCTS WHERE UNIT_PRICE BETWEEN 20000 AND 40000;

PROD_ID ITEM_NAME                     CAT_REF UNIT_PRICE QTY_IN_STOCK
------- ----------------------------- ------- ---------- ------------
P000005 Smartphone                    CAT004       28000           35
P000010 Tablet                        CAT004       22000           30
```

2. Display products belonging to a particular category.

```
SQL> SELECT * FROM PRODUCTS WHERE CAT_REF = 'CAT005';

PROD_ID ITEM_NAME                     CAT_REF UNIT_PRICE QTY_IN_STOCK
------- ----------------------------- ------- ---------- ------------
P000006 USB Cable                     CAT005         450          150
P000007 Power Bank                    CAT005        1800           75
```

3. Find products currently available in inventory.

```
SQL> SELECT * FROM INVENTORY WHERE AVAIL_STATE = 'Available';

INV_ID  PROD_ID VENDOR_ID QTY_ON_HAND AVAIL_STATE UPDATED_ON
------- ------- --------- ----------- ----------- ----------
INV0001 P000001 SEL0001            25 Available   15-JAN-26
INV0002 P000002 SEL0002           100 Available   20-JAN-26
INV0003 P000003 SEL0002            60 Available   25-JAN-26
INV0004 P000004 SEL0003            45 Available   05-FEB-26
INV0005 P000005 SEL0004            35 Available   15-FEB-26
INV0006 P000006 SEL0005           150 Available   01-MAR-26
INV0007 P000007 SEL0005            75 Available   10-MAR-26
INV0008 P000008 SEL0006            40 Available   20-MAR-26
INV0009 P000009 SEL0003            55 Available   05-APR-26
INV0010 P000010 SEL0004            30 Available   15-APR-26
INV0011 P000011 SEL0007            50 Available   01-MAY-26
INV0012 P000012 SEL0008            35 Available   15-MAY-26
INV0013 P000013 SEL0009            20 Available   10-JUN-26
INV0014 P000014 SEL0010            80 Available   15-JUL-26
INV0015 P000015 SEL0003            65 Available   20-SEP-26

15 rows selected.
```

4. Search products using product name.

```
SQL> SELECT * FROM PRODUCTS WHERE ITEM_NAME = 'Power Bank';

PROD_ID ITEM_NAME                     CAT_REF UNIT_PRICE QTY_IN_STOCK
------- ------------------------------ ------- ---------- ------------
P000007 Power Bank                    CAT005        1800           75
```

5. Find low-stock products.

```
SQL> SELECT * FROM INVENTORY WHERE QTY_ON_HAND < 30;

INV_ID  PROD_ID VENDOR_ID QTY_ON_HAND AVAIL_STATE UPDATED_ON
------- ------- --------- ----------- ----------- ----------
INV0001 P000001 SEL0001            25 Available   15-JAN-26
INV0013 P000013 SEL0009            20 Available   10-JUN-26
```

6. Retrieve Customer and Product Information

Tasks:

1. Display customer details with their orders.

```
SQL> SELECT
  2       C.CUST_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       C.EMAIL_ADDR,
  5       C.CONTACT_NO,
  6       O.ORDER_REF,
  7       O.ORDER_DATE,
  8       O.ORDER_TOTAL,
  9       O.ORDER_STATUS
 10  FROM CUSTOMERS C
 11  JOIN ORDERS O
 12    ON C.CUST_ID = O.CUST_ID
 13  ORDER BY C.CUST_ID, O.ORDER_DATE;

CUST_ID  CUSTOMER_NAME        EMAIL_ADDR                    CONTACT_NO ORDER_REF ORDER_DATE ORDER_TOTAL ORDER_STATUS
-------- -------------------- ----------------------------- ---------- --------- ---------- ----------- ------------
CUST0001 Arjun Kumar          arjun.kumar@gmail.com         9876543210 ORD0001   10-JAN-26        55850 Delivered
CUST0002 Priya Sharma         priya.sharma@gmail.com        9123456780 ORD0002   25-JAN-26         3350 Delivered
CUST0003 Rahul Menon          rahul.menon@gmail.com         9988776655 ORD0003   05-FEB-26        28000 Delivered
CUST0004 Ananya Iyer          ananya.iyer@gmail.com         9876123450 ORD0004   18-FEB-26         8300 Delivered
CUST0005 Vikram Reddy         vikram.reddy@gmail.com        9012345678 ORD0005   02-MAR-26        22000 Processing
CUST0006 Meera Krishnan       meera.krishnan@gmail.com      9345678120 ORD0006   15-MAR-26         7000 Delivered
CUST0007 Suresh Nair          suresh.nair@gmail.com         9567891234 ORD0007   01-APR-26         6500 Shipped
```

```
CUST_ID  CUSTOMER_NAME        EMAIL_ADDR                   CONTACT_NO ORDER_REF ORDER_DATE ORDER_TOTAL ORDER_STATUS
-------- -------------------- ---------------------------- ---------- --------- ---------- ----------- ------------
CUST0008 Kavya Rajan          kavya.rajan@gmail.com        9789012345 ORD0008   20-APR-26         5700 Delivered
CUST0009 Karthik Subramanian  karthik.subramanian@gmail.com 8899776655 ORD0009   05-MAY-26        18500 Processing
CUST0010 Divya Pillai         divya.pillai@gmail.com       9098765432 ORD0010   18-MAY-26         5200 Delivered
CUST0011 Aditya Verma         aditya.verma@gmail.com       9876501234 ORD0011   03-JUN-26         7000 Shipped
CUST0012 Sneha Patel          sneha.patel@gmail.com        9765432109 ORD0012   20-JUN-26         2450 Delivered
CUST0014 Lakshmi Venkatesan   lakshmi.venkatesan@gmail.com 9543210987 ORD0014   15-AUG-26        28000 Delivered
CUST0015 Nithin Murugan       nithin.murugan@gmail.com     9432109876 ORD0015   25-SEP-26         8350 Processing

14 rows selected.
```

2. Display product details with category information.

```
SQL> SELECT
  2       P.PROD_ID,
  3       P.ITEM_NAME,
  4       P.UNIT_PRICE,
  5       P.QTY_IN_STOCK,
  6       C.CAT_REF,
  7       C.CAT_NAME,
  8       C.CATEGORY_DESC
  9  FROM PRODUCTS P
 10  JOIN CATEGORIES C
 11    ON P.CAT_REF = C.CAT_REF
 12  ORDER BY C.CAT_NAME, P.ITEM_NAME;

PROD_ID ITEM_NAME                    UNIT_PRICE QTY_IN_STOCK CAT_REF CAT_NAME             CATEGORY_DESC
------- ---------------------------- ---------- ------------ ------- -------------------- ----------------------------
P000004 Bluetooth Speaker                  3200           45 CAT003  Audio Devices        Speakers headphones and earbuds
P000009 Headphones                         4200           55 CAT003  Audio Devices        Speakers headphones and earbuds
P000015 Wireless Earbuds                   3500           65 CAT003  Audio Devices        Speakers headphones and earbuds
P000003 Mechanical Keyboard                2500           60 CAT002  Computer Accessories Mouse and keyboard accessories
P000002 Wireless Mouse                      850          100 CAT002  Computer Accessories Mouse and keyboard accessories
P000014 Laptop Stand                       1500           80 CAT010  Laptop Accessories   Laptop stands and accessories
P000001 Laptop                            55000           25 CAT001  Laptops              Laptop computers and notebooks
P000005 Smartphone                        28000           35 CAT004  Mobile Devices       Smartphones and tablets
P000010 Tablet                            22000           30 CAT004  Mobile Devices       Smartphones and tablets
P000013 Gaming Monitor                    18500           20 CAT009  Monitors             Gaming and computer monitors
P000007 Power Bank                         1800           75 CAT005  Power Accessories    USB cables and power banks
P000006 USB Cable                           450          150 CAT005  Power Accessories    USB cables and power banks
P000012 External Hard Disk                 5200           35 CAT008  Storage Devices      External hard disks and storage
P000008 Smart Watch                        6500           40 CAT006  Wearables            Smart watches and wearable dev
P000011 Web Camera                         2800           50 CAT007  Web Accessories      Web cameras and related devices

15 rows selected.
```

3. Find customers who purchased a specific product.

```
SQL> SELECT DISTINCT
  2       C.CUST_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       C.EMAIL_ADDR,
  5       C.CONTACT_NO
  6  FROM CUSTOMERS C
  7  JOIN ORDERS O
  8    ON C.CUST_ID = O.CUST_ID
  9  JOIN ORDER_ITEMS OI
 10    ON O.ORDER_REF = OI.ORDER_REF
 11  JOIN PRODUCTS P
 12    ON OI.PROD_ID = P.PROD_ID
 13  WHERE P.PROD_ID = 'P000009';

CUST_ID  CUSTOMER_NAME        EMAIL_ADDR                    CONTACT_NO
-------- -------------------- ----------------------------- ----------
CUST0006 Meera Krishnan       meera.krishnan@gmail.com      9345678120
CUST0011 Aditya Verma         aditya.verma@gmail.com        9876501234
```

4. Display products purchased by each customer.

```
SQL> SELECT
  2       C.CUST_ID,
  3       C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4       O.ORDER_REF,
  5       P.ITEM_NAME,
  6       OI.ITEM_QTY,
  7       OI.UNIT_PRICE,
  8       OI.ITEM_QTY * OI.UNIT_PRICE AS LINE_AMOUNT
  9  FROM CUSTOMERS C
 10  JOIN ORDERS O
 11    ON C.CUST_ID = O.CUST_ID
 12  JOIN ORDER_ITEMS OI
 13    ON O.ORDER_REF = OI.ORDER_REF
 14  JOIN PRODUCTS P
 15    ON OI.PROD_ID = P.PROD_ID
 16  ORDER BY C.CUST_ID, O.ORDER_DATE, P.ITEM_NAME;

CUST_ID  CUSTOMER_NAME        ORDER_REF ITEM_NAME                    ITEM_QTY UNIT_PRICE LINE_AMOUNT
-------- -------------------- --------- ---------------------------- -------- ---------- -----------
CUST0001 Arjun Kumar          ORD0001   Laptop                              1      55000       55000
CUST0001 Arjun Kumar          ORD0001   Wireless Mouse                      1        850         850
CUST0002 Priya Sharma         ORD0002   Bluetooth Speaker                   1       3200        3200
CUST0002 Priya Sharma         ORD0002   USB Cable                           1        450         450
CUST0003 Rahul Menon          ORD0003   Smartphone                          6      28000      168000
CUST0004 Ananya Iyer          ORD0004   Smart Watch                         1       6500        6500
CUST0004 Ananya Iyer          ORD0004   USB Cable                           4        450        1800
```

```
CUST_ID  CUSTOMER_NAME        ORDER_REF ITEM_NAME                    ITEM_QTY UNIT_PRICE LINE_AMOUNT
-------- -------------------- --------- ---------------------------- -------- ---------- -----------
CUST0005 Vikram Reddy         ORD0005   Tablet                              1      22000       22000
CUST0006 Meera Krishnan       ORD0006   Headphones                          1       4200        4200
CUST0006 Meera Krishnan       ORD0006   Wireless Earbuds                    1       3500        3500
CUST0007 Suresh Nair          ORD0007   Smart Watch                         1       6500        6500
CUST0008 Kavya Rajan          ORD0008   External Hard Disk                  1       5200        5200
CUST0008 Kavya Rajan          ORD0008   USB Cable                           1        450         450
CUST0009 Karthik Subramanian  ORD0009   Gaming Monitor                      1      18500       18500
CUST0010 Divya Pillai         ORD0010   External Hard Disk                  1       5200        5200
CUST0011 Aditya Verma         ORD0011   Headphones                          1       4200        4200
CUST0011 Aditya Verma         ORD0011   Wireless Earbuds                    1       3500        3500
CUST0012 Sneha Patel          ORD0012   Laptop Stand                        1       1500        1500
CUST0012 Sneha Patel          ORD0012   Mechanical Keyboard                 1       2500        2500
CUST0015 Nithin Murugan       ORD0015   Wireless Earbuds                    1       3500        3500

20 rows selected.
```

7. Apply Multiple Filtering Conditions

Tasks:

1. Find electronics products costing more than ■10000.

```
SQL> SELECT
  2       P.PROD_ID,
  3       P.ITEM_NAME,
  4       C.CAT_NAME,
  5       P.UNIT_PRICE
  6  FROM PRODUCTS P
  7  JOIN CATEGORIES C
  8    ON P.CAT_REF = C.CAT_REF
  9  WHERE P.UNIT_PRICE > 10000
 10  ORDER BY P.UNIT_PRICE DESC;

PROD_ID ITEM_NAME                     CAT_NAME            UNIT_PRICE
------- ----------------------------- ------------------- ----------
P000001 Laptop                        Laptops                  55000
P000005 Smartphone                    Mobile Devices           28000
P000010 Tablet                        Mobile Devices           22000
P000013 Gaming Monitor                Monitors                 18500
```

2. Search products containing the word "Mobile".

```
SQL> SELECT
  2       PROD_ID,
  3       ITEM_NAME,
  4       UNIT_PRICE,
  5       QTY_IN_STOCK
  6  FROM PRODUCTS
  7  WHERE UPPER(ITEM_NAME) LIKE '%MOBILE%';

no rows selected
```

3. Find orders placed within a specific date range.

```
SQL> SELECT
  2       ORDER_REF,
  3       CUST_ID,
  4       ORDER_DATE,
  5       ORDER_TOTAL,
  6       ORDER_STATUS
  7  FROM ORDERS
  8  WHERE ORDER_DATE BETWEEN DATE '2026-03-01' AND DATE '2026-06-30'
  9  ORDER BY ORDER_DATE;

ORDER_REF CUST_ID  ORDER_DATE ORDER_TOTAL ORDER_STATUS
--------- -------- ---------- ----------- ------------
ORD0005   CUST0005 02-MAR-26        22000 Processing
ORD0006   CUST0006 15-MAR-26         7000 Delivered
ORD0007   CUST0007 01-APR-26         6500 Shipped
ORD0008   CUST0008 20-APR-26         5700 Delivered
ORD0009   CUST0009 05-MAY-26        18500 Processing
ORD0010   CUST0010 18-MAY-26         5200 Delivered
ORD0011   CUST0011 03-JUN-26         7000 Shipped
ORD0012   CUST0012 20-JUN-26         2450 Delivered

8 rows selected.
```

8. Generate Basic Business Reports

Report 1: Product Availability Report

Display:

- Product name

- Price
- Stock quantity
- Availability status

```
SQL> SELECT
  2       ITEM_NAME,
  3       UNIT_PRICE,
  4       QTY_IN_STOCK,
  5       CASE
  6            WHEN QTY_IN_STOCK > 0 THEN 'Available'
  7            ELSE 'Out of Stock'
  8       END AS AVAILABILITY_STATUS
  9  FROM PRODUCTS
 10  ORDER BY ITEM_NAME;

ITEM_NAME                                          UNIT_PRICE QTY_IN_STOCK AVAILABILITY_STATUS
-------------------------------------------------- ---------- ------------ ------------------
Bluetooth Speaker                                        3200           45 Available
External Hard Disk                                       5200           35 Available
Gaming Monitor                                          18500           20 Available
Headphones                                               4200           55 Available
Laptop                                                  55000           25 Available
Laptop Stand                                             1500           80 Available
Mechanical Keyboard                                      2500           60 Available
Power Bank                                               1800           75 Available
Smart Watch                                              6500           40 Available
Smartphone                                              28000           35 Available
Tablet                                                  22000           30 Available
USB Cable                                                 450          150 Available
Web Camera                                               2800           50 Available
Wireless Earbuds                                         3500           65 Available
Wireless Mouse                                            850          100 Available

15 rows selected.
```

Report 2: Customer Report

Generate:

- Total number of customers

```
SQL> SELECT COUNT(*) AS TOTAL_CUSTOMERS
  2  FROM CUSTOMERS;

TOTAL_CUSTOMERS
---------------
             15
```

- New customer registrations

```
SQL> SELECT
  2       CUST_ID,
  3       FIRST_NAME || ' ' || LAST_NAME AS CUSTOMER_NAME,
  4       EMAIL_ADDR,
  5       JOIN_DATE
  6  FROM CUSTOMERS
  7  WHERE JOIN_DATE >= DATE '2026-01-01'
  8    AND JOIN_DATE < DATE '2027-01-01'
  9  ORDER BY JOIN_DATE DESC;

CUST_ID  CUSTOMER_NAME        EMAIL_ADDR                    JOIN_DATE
-------- -------------------- ----------------------------- ---------
CUST0015 Nithin Murugan       nithin.murugan@gmail.com      03-OCT-26
CUST0014 Lakshmi Venkatesan   lakshmi.venkatesan@gmail.com  08-JUL-26
CUST0013 Rohan Bose           rohan.bose@gmail.com          17-APR-26
CUST0012 Sneha Patel          sneha.patel@gmail.com         20-FEB-26
```

Report 3: Order Report

Display:

- Total orders

```
SQL> SELECT COUNT(*) AS TOTAL_ORDERS
  2  FROM ORDERS;

TOTAL_ORDERS
------------
          14
```

- Completed orders

```
SQL> SELECT COUNT(*) AS COMPLETED_ORDERS
  2  FROM ORDERS
  3  WHERE ORDER_STATUS = 'Delivered';

COMPLETED_ORDERS
----------------
               9
```

- Pending orders

```
SQL> SELECT COUNT(*) AS PENDING_ORDERS
  2  FROM ORDERS
  3  WHERE ORDER_STATUS IN ('Processing', 'Shipped');

PENDING_ORDERS
--------------
             5
```

- Cancelled orders

```
SQL> SELECT COUNT(*) AS CANCELLED_ORDERS
  2  FROM ORDERS
  3  WHERE ORDER_STATUS = 'Cancelled';

CANCELLED_ORDERS
----------------
               0
```

Report 4: Product Performance Report

Find:

- Highest-priced products

```
SQL> SELECT
  2       PROD_ID,
  3       ITEM_NAME,
  4       UNIT_PRICE
  5  FROM PRODUCTS
  6  ORDER BY UNIT_PRICE DESC;

PROD_ID ITEM_NAME                                          UNIT_PRICE
------- -------------------------------------------------- ----------
P000001 Laptop                                                  55000
P000005 Smartphone                                              28000
P000010 Tablet                                                  22000
P000013 Gaming Monitor                                          18500
P000008 Smart Watch                                              6500
P000012 External Hard Disk                                       5200
P000009 Headphones                                               4200
P000015 Wireless Earbuds                                         3500
P000004 Bluetooth Speaker                                        3200
P000011 Web Camera                                               2800
P000003 Mechanical Keyboard                                      2500
P000007 Power Bank                                               1800
P000014 Laptop Stand                                             1500
P000002 Wireless Mouse                                            850
P000006 USB Cable                                                 450

15 rows selected.
```

- Most reviewed products

```
SQL> SELECT
  2       P.PROD_ID,
  3       P.ITEM_NAME,
  4       COUNT(R.FEEDBACK_ID) AS REVIEW_COUNT
  5  FROM PRODUCTS P
  6  JOIN PRODUCT_REVIEWS R
  7    ON P.PROD_ID = R.PROD_REF
  8  GROUP BY P.PROD_ID, P.ITEM_NAME
  9  ORDER BY REVIEW_COUNT DESC;

PROD_ID ITEM_NAME                                         REVIEW_COUNT
------- -------------------------------------------------- ------------
P000006 USB Cable                                                    3
P000008 Smart Watch                                                  2
P000012 External Hard Disk                                           2
P000015 Wireless Earbuds                                             2
P000009 Headphones                                                   2
P000014 Laptop Stand                                                 1
P000010 Tablet                                                       1
P000005 Smartphone                                                   1
P000001 Laptop                                                       1
P000004 Bluetooth Speaker                                            1
P000002 Wireless Mouse                                               1

11 rows selected.
```

- Available products

```
SQL> SELECT
  2       PROD_ID,
  3       ITEM_NAME,
  4       UNIT_PRICE,
  5       QTY_IN_STOCK
  6  FROM PRODUCTS
  7  WHERE QTY_IN_STOCK > 0
  8  ORDER BY QTY_IN_STOCK DESC;

PROD_ID ITEM_NAME                                          UNIT_PRICE QTY_IN_STOCK
------- -------------------------------------------------- ---------- ------------
P000006 USB Cable                                             450          150
P000002 Wireless Mouse                                        850          100
P000014 Laptop Stand                                         1500           80
P000007 Power Bank                                           1800           75
P000015 Wireless Earbuds                                     3500           65
P000003 Mechanical Keyboard                                  2500           60
P000009 Headphones                                           4200           55
P000011 Web Camera                                           2800           50
P000004 Bluetooth Speaker                                    3200           45
P000008 Smart Watch                                          6500           40
P000005 Smartphone                                          28000           35
P000012 External Hard Disk                                   5200           35
P000010 Tablet                                              22000           30
P000001 Laptop                                              55000           25
P000013 Gaming Monitor                                      18500           20

15 rows selected.
```